

ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

What place?
What time of day?
Does the place or time change?

ED reception/waiting area, Mercer General Hospital; continuous use, every shift.

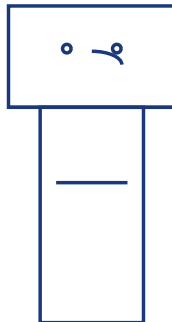
Robot's role

Is the robot a friend? Teacher? Helper?
Something else?

Patient-facing helper for self-check-in and data collection; not a diagnostician.

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?



Freestanding touchscreen kiosk with integrated vital-sign devices.

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?

Friendly and reassuring; uses simple plain language and calm audio/visual cues for first-time users.

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Adapts guidance using patient responses and live vitals; detects incomplete or abnormal input and escalates automatically.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Connects to Mercer EHR and nurse dashboard; sends ESI recommendations and vitals.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

☒ voice	☒ touch	☐ movement	☐ screens
☐ sounds	☐ smell	☐ lights	☒ other <u>barcode / ID scanner</u>
☐ gestures	☐ facial expressions		

OUTPUT

☒ voice	☐ touch	☐ movement	☒ screens
☒ sounds	☐ smell	☐ lights	☒ other <u></u>
☐ gestures	☐ facial expressions		

Interaction flow

Describe the most important interaction of the robot.

Note: only fill the bottom row if your robot is teleoperated.

ROBOT OPERATOR (optional)	ROBOT	USER	BEFORE	DURING	AFTER
			Patient arrives at reception and approaches the kiosk to self-check-in.	Patient follows on-screen/voice prompts to enter demographics and presenting complaint, and completes guided vital-sign readings.	Patient receives confirmation and waits, or is directed straight to a nurse if flagged.
			Kiosk displays a welcome screen and verifies patient identity.	Kiosk guides data entry, captures vitals from integrated devices, and generates a preliminary ESI recommendation.	Kiosk transmits data to the EHR/nurse dashboard and escalates automatically if issues are detected.

